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Artificial intelligence in school education: the case of Nigeria

L'intelligence artificielle dans l'éducation formelle : le cas du Nigéria

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Resúmenes

English Français

In line with global trends, the integration of artificial intelligence (AI) into the Nigerian school system has begun to take shape. However, efforts related to instructional and pedagogical design, professional learning programmes, and guidelines for best practices remain limited, given the inherent contextual features. Although the recently launched draft of the National AI Strategy shows promising development in building an AI ecosystem in the country, additional considerations are necessary to ensure the effective integration of AI into teaching and learning practices. This paper provides insights into teaching about AI and utilising AI in the classroom within the Nigerian school system. It further highlights the Nigerian government's initiatives on AI, including ethics, professional development needs, the state of instructional design, and the way forward for effective AI integration in the school system.

Au diapason des tendances internationales, l'intégration de l'intelligence artificielle (IA) dans le système scolaire nigérian a commencé. Cependant, les efforts en matière d'ingénierie didactique et pédagogique, les programmes de formation professionnelle et les orientations relatives aux bonnes pratiques demeurent limités, compte tenu du contexte du pays. Bien que le récent projet de Stratégie nationale pour l'IA montre une évolution prometteuse en matière de construction d'un écosystème de l'IA dans le pays, il est nécessaire de prendre en compte des éléments supplémentaires pour garantir l'intégration effective de l'IA dans les pratiques d'enseignement et d'apprentissage. Cet article traite de la manière dont l'IA est enseignée et utilisée en classe dans le système scolaire nigérian. Il souligne en outre les initiatives du gouvernement en matière d'IA, notamment en ce qui concerne les questions éthiques, les besoins en développement professionnel, l'état de l'ingénierie pédagogique et la voie à suivre pour une réelle intégration de l'IA dans le système scolaire.



Entradas del índice

Palabras claves : incertitude, intelligence artificielle, stratégie nationale, contrainte matérielle, développement professionnel

Keywords: uncertainty, artificial intelligence, national strategies, material constraints, professional development

Índice geográfico: Nigéria

Texto integral

- 1 Artificial intelligence (AI) increasingly impacts every facet of human life, including the education sector. Due to its revolutionary features, nations worldwide continue to find ways to harness AI's potential to improve their countries and educational systems. Nigeria, Africa's most populous country and sixth in world ranking with one of the largest youth populations, recognises the importance of AI in developing its demographic into a workforce that meets global AI ecosystem requirements. The recently launched draft of the National AI Strategy (NAIS, 2024) indicates Nigeria's interest in creating innovator pipelines and preparing individuals for an AI-driven future. While efforts have been made to transform Nigeria's digital economy – such as the creation of the Centre for Artificial Intelligence and Robotics (NCAIR) in 2020 (Oyelere *et al.*, 2022) –, initiatives relating to the application of AI in classrooms and the teaching of AI in school education remain limited. This article, with its dual emphasis on teaching AI and utilising AI in the classroom, provides insights into how AI has been integrated into the Nigerian school system. It offers an overview of Nigerian government initiatives on AI, including ethics, professional development needs, and instructional design. The paper concludes with recommendations for effectively integrating AI into the school system.

Government initiatives on AI in Nigeria

- 2 In November 2020, Nigeria launched the Centre for Artificial Intelligence and Robotics (NCAIR), a research facility focused on AI, robotics, drones, the Internet of Things, and other emerging technologies, with the aim of transforming the Nigerian digital economy (Oyelere *et al.*, 2022). In April 2024, the Nigerian government introduced the country's first multilingual large language model (LLM), designed to reflect its diversity and play a major role in the National AI Strategy. This new LLM was trained in five low-resource local languages and accented English, to ensure stronger language representation in existing datasets for developing AI solutions (NAIS, 2024).
- 3 In line with global standards, Nigeria introduced a draft of the National AI Strategy in August 2024, with the vision “to be a global leader in harnessing the transformative power of AI through responsible, ethical, and inclusive innovation, fostering sustainable development through collaborative efforts” (NAIS, 2024). In December 2024, the federal government further unveiled a free AI Academy¹ for Nigerian youth and civil servants to acquire cutting-edge AI skills. The AI Academy programme, officially starting in 2025, will provide participants with access to self-paced courses on AI fundamentals, ethics, tools, and real-world applications. Candidates were encouraged to register on the programme's website between mid-December and the end of January 2025. While AI initiatives continue to grow, challenges remain, as limited efforts are directed towards designing specific programmes for school education, including a limited focus in the K-12 educational level in the NAIS draft.

Current state of professional development

- 4 According to the NAIS draft, Nigeria's educational infrastructure is not yet robust enough to support widespread AI education. The document further notes that many schools lack the necessary resources – including technology and trained educators – to effectively teach AI-related courses. This is corroborated by a recent study investigating the factors influencing the use of AI by Nigerian science teachers (Nja *et al.*, 2023). The study found a high level of approval for using AI in science classes, but also highlighted concerns about inadequate training for teachers.
- 5 Sanusi *et al.* (2024) found that policymakers and teachers in Nigeria are concerned about teachers' AI competence. For example, a decision-maker stated that “teachers would be ready to mediate AI for their students if they were provided with sufficient training and had a good knowledge of AI.” A teacher also expressed that “teachers' knowledge of AI is important to help students develop their AI skills.” Their conclusions suggest building teachers' capacity through AI education programmes. The authors also recommend workshops and seminars to train in-service science teachers on the use of AI. Incorporating AI into teacher-training programmes would help teachers become familiar with AI early in their careers, thus building their confidence in using it. It was also recommended that guidelines on effectively using AI and its tools be made available to both trainee and in-service teachers.
- 6 Scientific evidence of initiatives to introduce AI curricula in schools – primarily through extracurricular programmes and after-school activities – remains limited in terms of implementation and reach. Nevertheless, these projects are generally led by NGO interventions and researchers' efforts to support students and teachers in learning AI. The Data Scientists Network is one such Nigerian NGO. It designs programmes such as boot camps and competitions to spark children's interest and prepare them for the opportunities offered by AI and data science. This organisation published AI textbooks for primary and secondary schools in Nigeria in 2019. Since 2020, researchers like Sanusi *et al.* (2024) have studied AI teaching for K-12 learners, with a strong emphasis on instructional design, student perspectives, and learner experiences in informal contexts.

Instructional design

- 7 The ongoing revolution in education, driven by the proliferation of AI, necessitates a rethinking of instructional design, which focuses on optimising the learning process. Achieving this optimisation requires attention to learning tasks, the design of learning materials, and the activation of learners' cognitive processes to invest effort into learning (Klepsch and Seufert, 2020). This indicates that fostering the construction of students' mental models requires rigorous design. Although some studies assert that coaching Nigerian learners to help them progressively acquire independent learning strategies has a significant impact on their problem-solving abilities, there is a lack of exemplary evidence on the design and implementation of scaffolding that supports self-regulated inquiry – with or without direct teacher or peer interaction – using AI. This is similarly true for teaching AI to school students in the region. Anecdotal evidence suggests that instructional design, student incentive and curriculum framework initiatives relating to learning AI are rarely studied.
- 8 Similarly, AI promotes autonomy in teaching and learning. Given the role of autonomy in fostering engagement and motivation, effective instructional design is crucial. The concept of autonomy – encompassing both learner and teacher autonomy – is particularly relevant in the context of AI in education. Learner autonomy involves the ability to take charge of one's learning with AI, while teacher autonomy refers to a

teacher's freedom to develop a personal pedagogy that balances their personality, training, experience, and the specific requirements of the educational context. There are studies on the effectiveness of metacognitive scaffolding to improve the agency and autonomy of learners in Nigerian schools. However, the specific use of AI to implement educational scaffolding is rarely highlighted in academic research or in practice.

- 9 While acknowledging academic integrity and ethical issues, the use of AI chatbots (generative AI) by students has been touted as a way to enhance engagement and autonomy by providing instant, tailored feedback and creating a dynamic learning environment. This fosters self-guided learning, empowering students to control their educational journey. However, there has been limited effort to harness and report on the effective integration of AI tools and applications in education. This calls for rigorous intervention alongside experimental and longitudinal education research, to evaluate how these tools can enhance education and improve the teaching–learning process, taking into account contextual features (Sanusi *et al.*, 2024).

Recommendations

- 10 While government initiatives continue to grow, many do not translate into classroom reality. The implementation of some policies is not followed through. For instance, according to TVC News², the federal government directed the Nigerian Educational Research and Development Council to integrate robotics and AI into the primary education curriculum. However, there is no evidence of its implementation. The government often issues directives without providing the necessary infrastructure and resources. This is corroborated by the NAIS draft, which states that many schools lack the resources – including technology and trained educators – to effectively teach AI-related courses.

- 11 There is a recognised need for research on instructional design for AI, yet only limited initiatives exist for teaching AI in Nigerian schools. Current efforts come from researchers exploring how to support learning for K-12 students through informal contexts and from gathering teachers' and policymakers' perspectives on promoting AI in schools (Sanusi *et al.*, 2024). Currently, the use of AI in classrooms or the teaching AI in schools is not widespread in Nigeria. Due to the uneven distribution of resources, some privileged public schools, top-rated private schools, or mission schools with abundant resources may teach AI or adopt AI tools to support their teaching and learning processes. A teacher from a prestigious secondary school in Nigeria's most cosmopolitan city told us:

In my school, we have a robotics class that allows us to present AI basics and projects to our students, including in the lower levels.³

- 12 Conversely, a science teacher encountered during a professional development course that we facilitated informed us that his school had only just been equipped with an IT lab, provided by former students. These personal experiences highlight the concerns raised in Sanusi *et al.*'s study (2024), which stated that, without current measures,

it might just be that some private schools will include AI as part of their subjects and the public schools' students would not be able to access AI content.

- 13 Given that nearly half of Nigeria's population is rural, a significant number of students and teachers do not benefit from the global AI revolution and may be ill-prepared for future skills. To address the challenge of uneven access to AI education, the government should deploy more resources to schools in low-resource settings. A public–private partnership model could also be adopted, leveraging international organisations, non-governmental organisations, and partnerships with tech companies.

- 14 There is also a need to create funding facilities and instruments to support researchers and instructional designers in providing evidence of implementable

learning materials, tools, and resources to ensure effective integration in schools. Curriculum developers should revise the current school curricula to accommodate AI tools, encouraging teachers to adopt these requirements. Teachers, instructional designers, and relevant stakeholders should be involved in designing AI learning experiences by creating a community where they co-design and prototype learning materials and activities that support AI integration across the curriculum. The national strategy, as currently drafted, only vaguely mentions initiatives to introduce AI curricula in schools (NAIS, 2024, p. 32) and lacks clear policy direction for AI education and how to achieve AI education for all.

- 15 While ensuring responsible and ethical AI development is one of the pillars of the NAIS draft, the policy should translate this into classroom reality, making teachers and students aware of the responsible use of AI and the ethics of the AI they encounter daily. More rigorous research is needed to understand how to effectively integrate AI applications in the classroom and how to teach K-12 learners about AI. Policymakers should consider adopting research findings to improve AI integration in the Nigerian school system.

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Notas

1 <https://guardian.ng/news/fg-unveils-free-ai-academy/>

2 <https://www.tvcnews.tv/fg-directs-nerdc-to-incorporate-robotics-ai-into-educational-curricula/>

3 Conversation with the author, February 8, 2025.

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